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SULTAN IDRIS EDUCATION UNIVERSITY

Software Verification and Validation (DES3103)

by

Dr. Nor Asiah Mohamad@Razak

Course Learning Outline

1. Briefing Session (Ice Breaking via Whiteboard, Class rules & Pro forma): **M. Teams**
2. Introduction to Software Verification & Validation I & II (V&V)
3. Managing V&V with SQA Integration
4. Quality metrics or attributes in V&V

WEEK 6-MID-TERM TEST (QUIZ): Topic 1 – 3-(CLO1:C2) – 20 MARKS

WEEK 7-PHASE 1: ASSIGNMENT 1-ONLINE PROJECT-BASED LEARNING (OPBL)-(CLO3:A5) - 20 MARKS: M. Teams

WEEK 9-PHASE 1: PRESENTATION & REFLECTION ON PHASE 1: M. Teams

5. Software Reviews – Will be covered on Assignments 1 & 2
6. Quality Control: Software Testing - Will be covered in Assignments 1 & 2
7. Human-Computer Interface Testing - Will be covered in Assignments 1 & 2
8. Manual Testing and Automation Testing: Will be covered in Final Assessment

WEEK 11-PHASE 2: ASSIGNMENT 2-ONLINE PROJECT-BASED LEARNING (OPBL)-(CLO2&4:P4&C5) - 20 MARKS: M. Teams
WEEK 13-PHASE 2: PRESENTATION & REFLECTION ON PHASE 2: M. Teams

9. Quality Maturity Models
10. Software Problem Analysis and Reporting
11. Quality Culture in doing V&V.
12. Safety and Security in V&V

WEEK 14: REFLECTION ON IMPLEMENTATION OPBL (SOFTWARE TESTERS) IN THE V&V COURSE: e-FORUM

WEEK 16-PHASE 3: FINAL ASSESSMENT – INDIVIDUAL SPECIAL PROJECT BASED ON PHASE 2 (CLO4:C5) (2 WEEKS) - 40 MARKS : e-ASSESSMENT

TOPIC 3

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Managing V&V with Integrating SQA

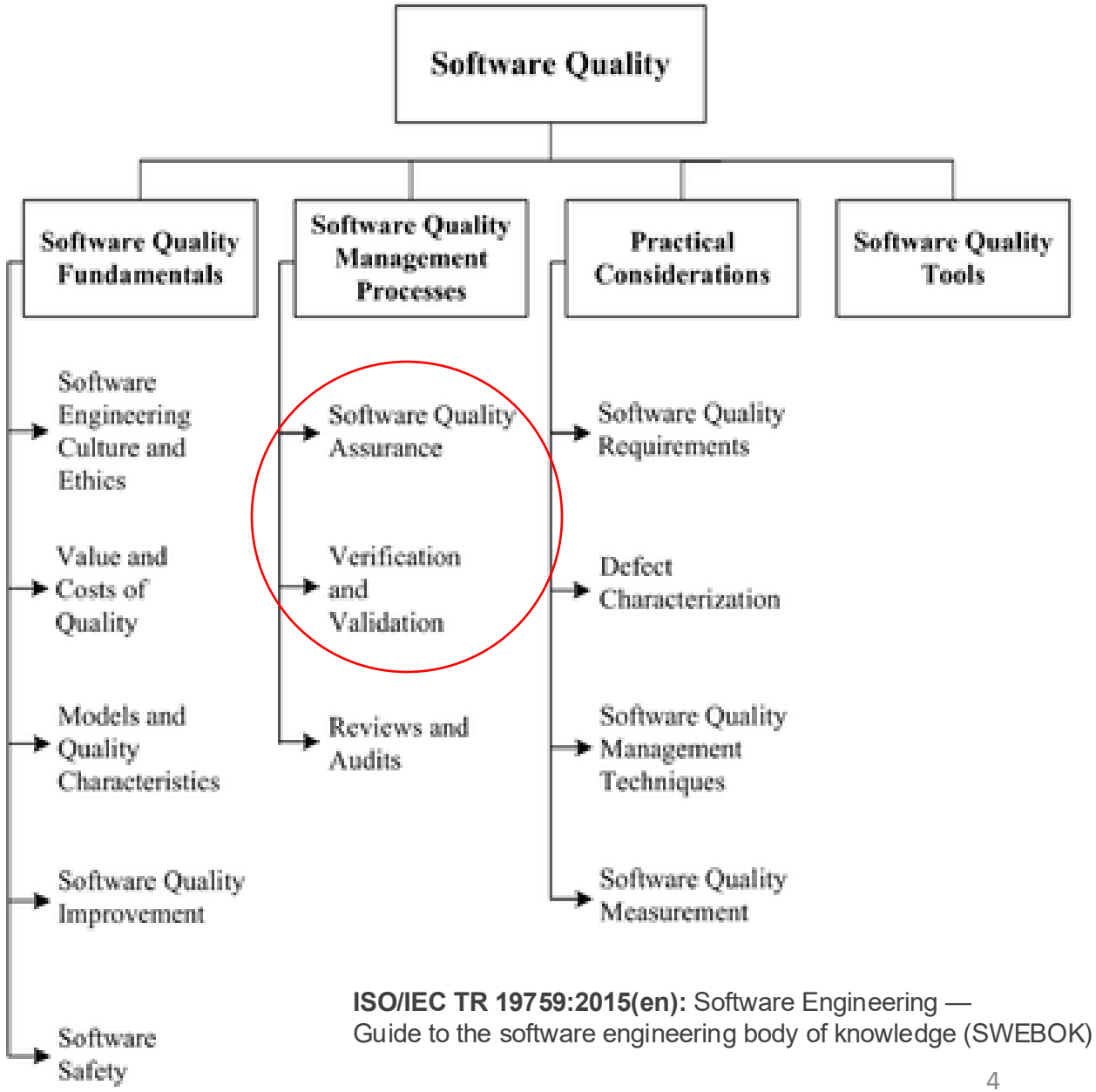
- Managing V&V by Integrating SQA in the software project lifecycle
- Quality issues and problems in the software development process and fitness among SQA Quality management
- Quality control in the software development process

Managing V&V by Integrating SQA

- Managing V&V with SQA Integrating SQA in software project lifecycle

Where will I see SQA and how will I recognize it?

- SDLC & Quality Assurance



ISO/IEC TR 19759:2015(en): Software Engineering — Guide to the software engineering body of knowledge (SWEBOK)

ISO/IEC Std. 12207-2008: system and software engineering – software life cycle processes

ISO/IEC/IEEE Std. 12207-2008 (ISO/IEC, 2008) provides a framework that incorporates the entire spectrum of software life cycle processes.

The objectives of the 12207-2008 Standard can be summarized thus:

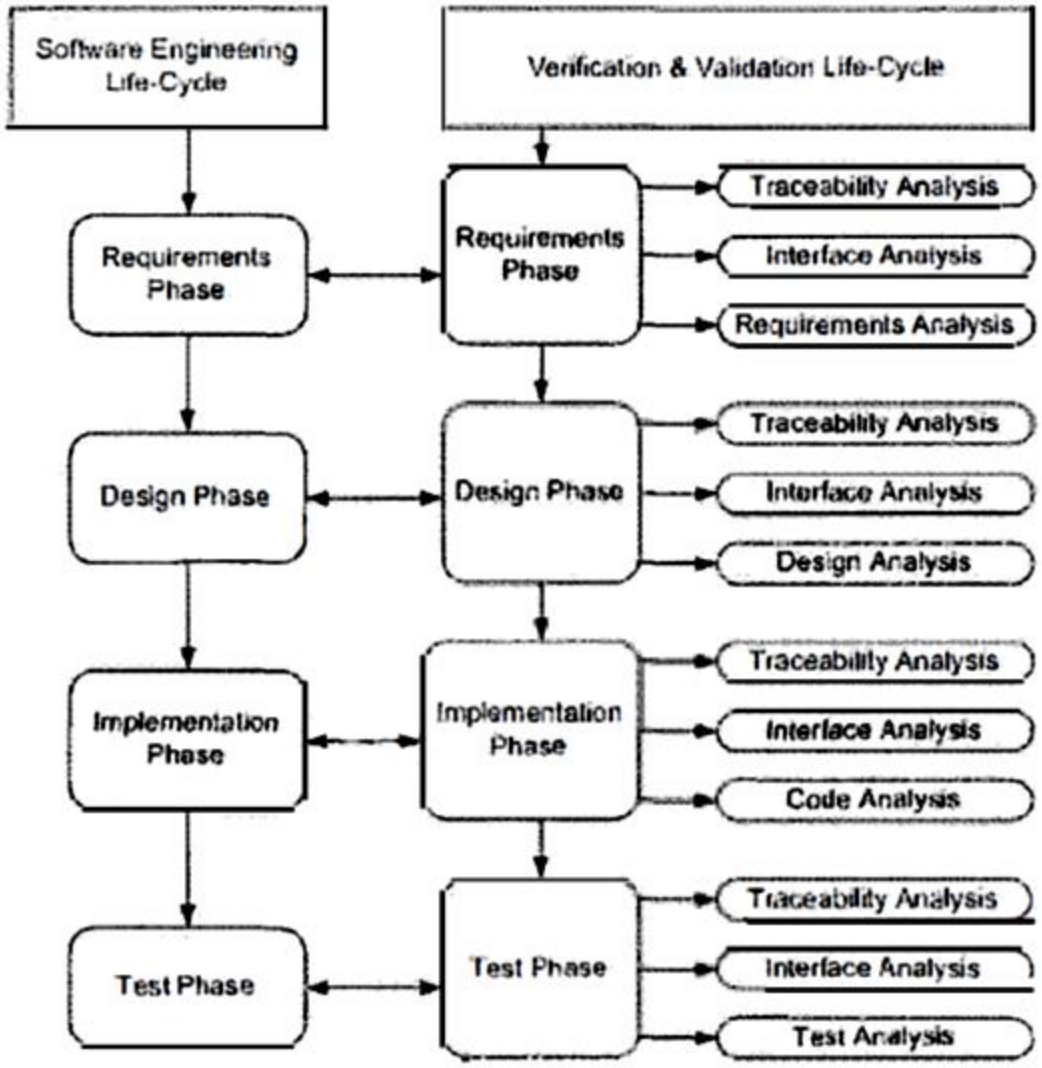
- To establish an internationally recognized model of common software life cycle processes that can be referenced by the software industry worldwide to facilitate communication among acquirers, suppliers, and other stakeholders.
- To serve all participants in a software life cycle: acquirers of software and system products, suppliers, developers, operation and maintenance staff, managers, SQA staff, and product users.
- To promote understanding among business parties by applying commonly recognized processes, activities, and tasks.

The concepts expressed in this standard deal with applicability to every participant in the software life cycle. The standard applies to all participants in the software life cycle: buyers, suppliers, developers, operators, and maintenance professionals. It provides separate definitions of processes, activities, and tasks for each role.

Managing V&V with SQA Integrating SQA - Managing V&V with SQA Integrating SQA in software project lifecycle

V&V Life -Cycle

- A software system evolves through a process that we traditionally call a life-cycle. It begins with a definition that describes what behavior it is supposed to have. A solution is then created that is supposed to reflect that behavior. This solution is then built and tested against the initial behavioral description. If it passes then it gets deployed, if it doesn't then it gets reworked until it does pass.
- These artifacts represent the system at particular stages of its life.
- V&V is itself a process and has a life-cycle of its own.
- The V&V lifecycle is run in parallel with the developers.
- For example, as the behavior is defined, and its byproduct generated (e.g., requirements specification) V & V will perform requirements analysis.
- Based on their assessment V & V will gain an understanding of the system's behavior, will generate specific facts regarding the quality of the requirements, and may generate issues with the documented requirements.
- The figure depicts the life-cycle that V&V traditionally follows. The figure shows the relationships with the software engineering life-cycle.



Traditional life-cycle that a verification and validation (V&V) project will follow. During development, the software evolves through a life-cycle and the V&V project is ran in parallel.

Managing V&V with SQA Integrating SQA – Characteristics of Software Quality Attributes

FIVE SOFTWARE QUALITY ATTRIBUTES

Atribut	Keterangan	Contoh
Atribut fungsian (*Functional attributes*)	Input dan output produk perisian	i. Sistem dapat memenuhi keperluan pengguna dengan tepat (*correctness*) . ii. Sistem mengawal capaian modul mengikut peranan pengguna (*authentication*). iii. Sistem memaparkan maklumat sulit kepada pengguna tertentu sahaja (*integrity*).Aspek keselamatan telah diambil berat semasa pembangunan sistem (*security*)
Atribut operasi (*Operational attributes*)	Syarat operasi sesuatu produk perisian	i. Sistem berupaya memberikan tindak balas (*latency or response time*) dalam masa yang ditetapkan ii. Sistem dapat menampung kapasiti (*capacity*) pengguna serentak yang ditetapkan iii. Sistem dapat beroperasi walaupun melebihi kapasiti pengguna yang ditetapkan.
Atribut kebolehgunaan (*Usability attributes*)	Sejauh mana produk perisian boleh digunakan dan disesuaikan dengan produk perisian	i. Sistem mudah digunakan (*ease of use*) oleh pengguna ii. Sistem mudah dipelajari (*ease of learn*) oleh pengguna iii. Sistem mudah disesuaikan (*customizable*) mengikut keperluan operasi pengguna iv. Sistem sedia berkolaborasi dengan aplikasi lain (*interoperability*)
Atribut perniagaan (*Business attributes*)	Kos pembangunan, penggunaan serta perubahan produk perisian	i. Kos pembangunan sistem ii. Kos penyelenggaraan system iii. Penjimatan kos apabila produk atau komponennya boleh diguna semula (*reusability*) iv. Penjimatan kos apabila produk boleh digunakan pada platform berbeza (*portability*)
Atribut struktur (*Structural attributes*)	Struktur dalaman produk perisian	i. Sistem yang dibangunkan berupaya untuk diuji (*testability*) ii. Sistem mudah diadaptasi (*adaptability*) mengikut perubahan keperluan pengguna iii. Pembangunan sistem mengambil kira struktur kemodularan (*modularity*)

Source: KRISA Methodology, Malaysian Administrative Modernisation and Management Planning Unit (MAMPU)

Managing V&V with SQA Integrating SQA - Quality issues and problems in the software development process and fitness among SQA Quality management

Nine major causes of software errors

- Faulty requirements definition
- Business – IT communication failures
- Deliberate deviations from software requirements
- Logical design errors
- Coding errors
- Non-compliance with documentation and coding instructions
- Shortcomings of the testing process
- User interface procedure errors
- General documentation errors

Source: Software Quality Assurance by Daniel Galin

Managing V&V with SQA Integrating SQA - Quality issues and problems in the software development process and fitness among SQA Quality management

1. **Faulty Requirements Definition**
   - **Usually considered the root cause of software errors**
   - **Incorrect requirement definitions**
     - **Simply stated, 'wrong' definitions (formulas, etc.)**
   - **Incomplete definitions**
     - **Unclear or implied requirements**
   - **Missing requirements**
     - **Just flat-out 'missing.' (e.g. Program Element Code)**
   - **Inclusion of unneeded requirements**
     - **(many projects have gone amuck for including far too many requirements that will never be used.**
     - **Impacts budgets, complexity, and development time**

Source: Software Quality Assurance by Daniel Galin

Managing V&V with SQA Integrating SQA - Quality issues and problems in the software development process and fitness among SQA Quality management

2. **Client-developer communication failures**
   - **Misunderstanding of instructions in requirements documentation (written / graphical instructions)**
   - **Misunderstanding of written <u>changes</u> during development.**
   - **Misunderstanding of <u>oral</u> changes during development.**
   - **Lack of attention**
     - **to client <u>messages</u> by developers dealing with requirement changes and**
     - **to client <u>responses</u> by clients to developer questions**

- **Very often, these very talented individuals come from different planets, it seems.**
  - **Clients represent the users; developers represent a different mind set entirely some times!**

Source: Software Quality Assurance by Daniel Galin

Managing V&V with SQA Integrating SQA - Quality issues and problems in the software development process and fitness among SQA Quality management

3. **Deliberate deviations from software requirements**
   - **Developer reuses previous / similar work to save time.**
   - **Often reused code needs modification which it may contain contain unneeded / unusable extraneous code.**
- **Book suggests developer(s) may overtly <u>omit</u> functionality due to time / budget pressures.**
  - **Another BAD choice; System testing will uncover these problems to everyone's dismay!**
  - **I have never seen this done intentionally – but understand it!**
- **Developer inserting unapproved 'enhancements' (perfective coding; a slick new sort / search….); may also ignore some seemingly minor features, which sometimes are quite major.**
  - **Have seen this and it too causes problems and embarrassment during reviews.**

Source: Software Quality Assurance by Daniel Galin

Managing V&V with SQA Integrating SQA - Quality issues and problems in the software development process and fitness among SQA Quality management

4. Logical design errors

- Definitions that represent software requirements by means of erroneous algorithms.
  - Yep! Wrong formulas;
  - Wrong Decision Logic Tables;
  - Incorrect descriptions in text
- Process definitions: procedures specified by systems analyst not accurately reflecting the real business process
  - Note: all errors are not necessarily software errors.
  - This seems like a procedural error, and likely not a part of the software system…But they are errors nonetheless!
- Erroneous Definition of Boundary Condition – a common source of errors
  - The "absolutes" like 'no more than' "fewer than," "n times or more;" "the first time," etc.

Source: Software Quality Assurance by Daniel Galin

Managing V&V with SQA Integrating SQA - Quality issues and problems in the software development process and fitness among SQA Quality management

4. Logical design errors

- **Omission of required software system states**
  - **If rank is >= O1 and RPI is numeric, then….easy to miss an action based on the software system state.**
- **Omission of definitions / procedures concerning reactions to illegal operation of the software system.**
  - **Including not only code to detect an illegal operation but failure to design the computer software reaction to this: Gracefully terminate, sound alarm, etc. ??**

Source: Software Quality Assurance by Daniel Galin

Managing V&V with SQA Integrating SQA - Quality issues and problems in the software development process and fitness among SQA Quality management

5. Coding errors

- Too many to try to list.
  - Syntax errors (grammatical errors)
  - Logic errors (program runs; results wrong)
  - Run-time errors (crash during execution)

- Know the differences. Be also able to supply an example. Examples do not 'define' something, but they may be used to 'support' or 'supplement' a definition.

Source: Software Quality Assurance by Daniel Galin

Managing V&V with SQA Integrating SQA - Quality issues and problems in the software development process and fitness among SQA Quality management

6. Non-compliance with documentation and coding instructions
   - Non-compliance with published templates (formats)
   - Non-compliance with coding standards
     - (Standards and Integration Branch)
     - Size of program;
       - Other programs must be able to run in environment!
     - Coding: Data Elements and Codes: AFM 300-4;
     - Development Procedures: Required documentation manuals and operating instructions; AFDSDCM 300-8, etc…
   - SQA Team: testing not only execution software but coding standards; manuals, messages displayed; resources needed; resources named (file names, program names,…)

Source: Software Quality Assurance by Daniel Galin

7. Shortcomings of the Testing Process

- **Likely the part of the development process cut short most frequently!**
- **Incomplete test plans**
  - **Parts of application not tested or tested thoroughly!**
  - **Superficial; boundary conditions…**
  - **Path testing, branch testing … (coverage measures)**
- **Failure to document and report detected errors and faults**
  - **So many levels of testing….we will cover.**
- **Failure to quickly correct detected faults due to unclear indications that there 'was' a fault**
- **Failure to fix the error due to time constraints**
  - **Many philosophies here depending on severity of error.**

Source: Software Quality Assurance by Daniel Galin

Managing V&V with SQA Integrating SQA - Quality issues and problems in the software development process and fitness among SQA Quality management

8. **User Interface and Procedure errors**

   Remember: to the user, the interface is the entire system.

   If the Interface is unsatisfactory, this view will be absolutely conveyed 'up the line.'

   The 'learnability,' and utility of the interface.

9. **Documentation Errors**
   - **Errors in the design documents**
     - **If Docs do not represent the implemented design, this is trouble for subsequent redesign and reuse**
   - **Errors in the documentation in the User Manuals, Operators Manual, other manuals (Installation…)**
   - **Errors in on-line help, if available.**
   - **Listing of non-existing software functions**
     - **Planned early but dropped; remain in documentation!**

- **Many error messages are totally meaningless**

Source: Software Quality Assurance by Daniel Galin

IEEE Std. 1012-2012 concepts

The concepts expressed in IEEE 1012-2012 address six basic issues:

1. **Broad definition of V&V activities**
   Broad definition of V&V activities enables the standard to embrace all SQA activities performed throughout the software life cycle, relating to management, systems, software, and hardware.

2. **Integrity levels** system performance
   "Major" – A function that affects important system performance.
   "Moderate" – A function that affects system performance; however, availability of an alternative method of operation enables the system to overcome the associated difficulties.
   "Low" – A function that affects system performance only by inconveniencing the user.
   The standard requires that integrity levels be assigned to components as early as the software V&V plan (SVVP) is prepared.

3. **Minimum V&V tasks**
   The standard defines the minimum tasks required for each integrity level and includes tables of optional task selection for each integrity level.

IEEE Std. 1012-2012 concepts

The concepts expressed in IEEE 1012-2012 address six basic issues:

4. **Detailed criteria for V&V tasks**
   The standard includes criteria for each V&V task, including minimum criteria for correctness, consistency, completeness, accuracy, readability, and testability. The V&V task descriptions include a list of required inputs and outputs.
5. **Intensity and rigor applied to V&V tasks**
   According to the standard, the intensity and rigor applied to the V&V tasks vary according to the integrity level, the higher the integrity level, the higher the required intensity and rigor applied to the V&V task.
6. **Detailed criteria for V&V tasks**
   The V&V task descriptions include a list of required inputs and outputs. They also include specific quantitative criteria for each V&V task, including minimum criteria for correctness, consistency, completeness, accuracy, readability and testability.
7. **Compliance and compatibility to international standards**
   The IEEE Std. 1012-2012 standard defines the V&V processes to conform to the international life cycle standards ISO/IEC/IEEE Std. 12207-2008 and ISO/IEC/IEEE Std. 15288-2008, as well as to the entire group of IEEE typical software engineering standards.

Managing V&V with SQA Integrating SQA - Quality control in software development process

- The implementation of software quality control is a much more recent development.
- As the Information Age dawned in the mid-1900s, it became apparent that quality control standards should be applied to the development of software, just like the manufacturing of physical products.
- Though digital and physical products are quite dissimilar, the application of quality control standards shares a common goal to both: the production of products that consistently serve users as designed.

Differences Between Quality Control and Quality Assurance

The terms 'quality control' and 'quality assurance' are often used interchangeably, but they each encapsulate separate tasks and activities. While quality assurance focuses more on ensuring quality during the software development process, quality control is more about identifying defects and flaws in completed software products. Thus, quality assurance is more prevention-oriented, while quality control is more detection-oriented.

The scope of the two are also different. Software quality assurance tends to be more organizational in breadth, focusing on all software development efforts that occur within an enterprise. Software quality control is comprised of a set of activities that focus upon monitoring the quality of a single software product — though likely duplicated across an organization's entire suite of software products.

Software quality assurance is used during most phases of the Software Development Life Cycle, quality control is typically restricted to the review and testing phases. However, there is one common activity that's integral to both quality assurance and quality control: testing.

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Thank You…
Terima kasih..